

Stuti Shah

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EDUCATION

Virginia Tech <i>Master of Engineering in Computer Science</i>	Blacksburg, VA Aug 2025 – May 2026
Virginia Tech <i>Bachelor of Science in Computer Science, Minor in Economics</i>	Blacksburg, VA Aug 2022 – May 2025

EXPERIENCE

Software Engineering / Data Science Intern <i>Volvo Group</i>	May 2025 – Aug 2025 <i>Greensboro, NC</i>
- Built backend and analytics services using Python, Flask, PostgreSQL, Docker, Kubernetes, and Argo CD, improving reliability, deployment reproducibility, and maintainability across engineering workflows. - Automated DMIS output file generation using Streamlit, reducing processing time from 4 hours to under 6 minutes and delivering \$176,980 in annual operational savings across distributed engineering teams. - Designed an end-to-end ML analytics platform linking complete vehicle modeling signals with millions of vehicle sensor records for scalable predictive performance analysis and engineering decision support. - Developed REST APIs to connect high-volume vehicle test data with analytics pipelines, improving data traceability, query access, and workflow execution speed. - Built and validated Random Forest models to predict truck component damage and acceleration, supporting data-driven test readiness decisions across vehicle platforms and configurations. - Integrated predictive models into analytics workflows, including component performance prediction and GHG emissions estimation.	
AI Intern <i>AquaOrange</i>	Jun 2024 – Aug 2024 <i>Bangkok, Thailand</i>
- Automated test workflows for 10+ Zoho CRM customizations using BrowserStack and testRigor, improving regression coverage across frontend and backend features. - Built validation pipelines for customized CRM workflows, reducing manual QA effort and improving defect detection during release cycles.	
Data Science Intern / Solutions Associate <i>HERE! Wireless USA</i>	Jun 2023 – Jul 2024 <i>Dallas, TX</i>
- Executed 50+ alpha testing workflows across Android and iOS applications, improving parental control feature stability and reducing post-release issues. - Conducted root-cause analysis for mobile app issues, helping improve feature reliability and user experience across product iterations. - Collaborated with product managers and engineers to translate customer feedback into actionable product and engineering requirements. - Developed four pricing models to support leadership in minimizing revenue leakage, improving market positioning, and optimizing product strategy.	
CS Undergraduate Teaching Assistant <i>Virginia Tech</i>	Aug 2024 – May 2025 <i>Blacksburg, VA</i>
Mentored 200+ students in computer systems topics including processor architecture, assembly programming, memory hierarchy, caching, and performance optimization.	

PROJECTS

Multimodal Uncertainty-Aware Stock Return Forecasting After Earnings Calls
- Built a multimodal forecasting pipeline for next-trading-day post-earnings log returns using S&P 500 earnings call transcripts, structured financial indicators, and Reddit-based sentiment features. - Implemented a PyTorch cross-modal fusion model with multi-head attention and conformalized quantile regression for calibrated interval prediction.
YOLO11 Shelf Detection for Planogram Compliance

- Built an end-to-end shelf-auditing system using YOLO11, OpenCV, Qwen2-VL, Flask, and AWS EC2 to detect products, identify empty shelf spaces, and assess planogram compliance.

CS Source JobHub

- Developed backend services and REST APIs using Flask, SQLAlchemy, and MySQL to support employer registration, contact management, and membership workflows.

TECHNICAL SKILLS

Programming: Python, Java, C, SQL

AI & ML: PyTorch, scikit-learn, TensorFlow, model evaluation, feature engineering

Backend & Systems: Flask, REST APIs, Linux, PostgreSQL, MySQL, SQLAlchemy

Cloud & DevOps: AWS EC2, S3, Redshift, SageMaker, Docker, Kubernetes, Argo CD, GitOps, CI/CD

Analytics: Spark/PySpark, Streamlit, Power BI, Tableau, ETL pipelines